

TERMINAL VELOCITIES AND THE BISTABILITY OF STELLAR WINDS

HENNY J. G. L. M. LAMERS,¹ THEODORE P. SNOW,² AND DOUGLAS M. LINDHOLM²

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ABSTRACT

The terminal velocity, v_∞ , of the stellar winds of 117 stars of spectral types O through F has been measured from the spectra in the new *Atlas of Ultraviolet P Cygni Profiles* by Snow et al. For the 68 stars with the most accurate determination of v_∞ we studied the relation between v_∞ and the photospheric escape velocity, v_{esc} . The ratio v_∞/v_{esc} decreases from 2.58 ± 0.20 in O-stars to 0.72 ± 0.15 in A and F stars. There is a clear discontinuity in this ratio near $T_{\text{eff}} \simeq 21,000$ K at spectral type B1, where the ratio drops by almost a factor 2 from $v_\infty/v_{\text{esc}} \simeq 2.6$ at the high-temperature side $v_\infty/v_{\text{esc}} \simeq 1.3$ at the low-temperature side, and possibly a smaller discontinuity near $T_{\text{eff}} \simeq 10,000$ K. The discontinuity near 21,000 K is due to the change in the nature of the lines that produce the radiation pressure. At $T_{\text{eff}} \gtrsim 21,000$ K the radiation pressure is due to high-ionization lines which are largely optically thick, whereas at $T_{\text{eff}} \lesssim 21,000$ K the radiation pressure is due to very large numbers of low-ionization lines, which are more optically thin. Near $T_{\text{eff}} \simeq 21,000$ K the winds can exist in two states: winds with high mass-loss rates and low velocities, and winds with low mass-loss rates and high velocities. This is the *bistability* of stellar winds for stars with $T_{\text{eff}} \simeq 21,000$ K.

We present evidence that the bistability may produce a jump in the mass-loss rates of the stars near $T_{\text{eff}} \simeq 21,000$ K with the higher mass-loss rates at the low temperature side of the jump. We briefly discuss the possible role of bistability in the formation of disks around B[e]-supergiants.

The observed values of v_∞ are used to derive empirical values of the force multiplier parameter α_{emp} that can be used to predict the terminal velocity of any early-type star with an accuracy of better than 5% with the method described in Appendix B, if the stellar parameters are known.

Subject headings: stars: early-type — stars: mass loss — supergiants — ultraviolet: stars

1. INTRODUCTION

The winds of early-type stars (types O to F) are driven by radiation pressure in spectral lines. If the radiation pressure is known, the mass-loss rates and the terminal velocities of the winds can be predicted. Alternatively the observed terminal velocities of the stellar winds can be used to derive information about the radiation pressure. This is the purpose of this study.

The radiation pressure due to spectral lines can be expressed in terms of three “force multiplier parameters,” k , α , and δ (Castor, Abbott, & Klein 1976, hereafter CAK; and Abbott 1982). These coefficients parameterize the radiative acceleration, g_{rad} , i.e., the gradient of the radiation pressure, as

$$g_{\text{rad}} \sim k(dv/dr)^\alpha g_e, \quad (1)$$

where $g_e = \sigma_e L/4\pi c R_*^2$ is the radiative acceleration by electron scattering. The parameter k is dependent on the number of lines that produce the radiation pressure, and the parameter α depends on the optical depth of the driving lines. If all the driving lines were optically thick then α would be 1. If all the driving lines were optically thin then α would be 0. In reality the radiation force is due to a mixture of optically thick and thin lines and so $0 < \alpha < 1$. The force multiplier parameter δ describes the dependence of k on the density with $k \sim \rho^\delta$. The velocity law of a radiation-driven wind depends on α and δ , but not on k . The mass-loss rate depends on k , α , and δ (CAK; Abbott 1982; Pauldrach, Puls, & Kudritzki 1986).

The values of k , α , and δ have been calculated in non-LTE by Pauldrach et al. (1990) for O-stars. The values were calculated

for a wider temperature range of $6000 \leq T_{\text{eff}} \leq 50,000$ K by Abbott (1982). He used LTE calculations for more than 10^5 lines for a coarse grid of temperatures and found that α drops from ~ 0.60 at $T_{\text{eff}} \gtrsim 30,000$ K, to 0.51 at $8000 \lesssim T_{\text{eff}} \lesssim 20,000$ K and to 0.46 at $T_{\text{eff}} \simeq 6000$ K. This results in stellar wind models in which the ratio v_∞/v_{esc} slowly decreases with decreasing T_{eff} , because $v_\infty/v_{\text{esc}} \sim [\alpha/(1-\alpha)]^{1/2}$.

However, Pauldrach & Puls (1990) have shown in a theoretical study of the wind of the Luminous Blue Variable P Cygni (B1 Ia⁺, $T_{\text{eff}} = 19,300$ K) that the wind may be bi-stable if the optical depth of the wind in the Lyman continuum is close to unity. In that case there are two solutions for the radiation-driven wind. In one solution the ionization is “high” and α is relatively large so the radiation pressure produces a “high” terminal velocity of ~ 600 km s^{−1} and a relative low density in the wind. If the temperature drops a little (by less than 1000 K) the wind will recombine to lower ionization stages which produces a smaller value of α and hence a “low” terminal velocity of ~ 200 km s^{−1} and a relatively high density in the wind.

This raises the interesting question: is this bistability mechanism limited to variable stars or can it also be detected in normal stars as a sudden change in the structure of the wind between stars that are on the “hot side” of the limit and stars that are on the “cool side” of the limit?³ And if this is the case, does it only result in a discontinuous distribution in *wind velocity* or is there also a jump in *mass-loss rate* at that particular value of T_{eff} ?

³ The term *bistability* suggests that the stellar winds can change between two states. This is not necessarily the case here, where we compare stars in one state with other stars in the other state. However, we will adopt the same nomenclature because the mechanism that produces the difference between the two states of different stars may also produce real bistability in some special cases, e.g., in P Cygni.

¹ SRON Laboratory for Space Research and Astronomical Institute, Utrecht, The Netherlands.

² Center for Astrophysics and Space Astronomy, University of Colorado, Boulder, CO 80309-0389.

To answer these questions we studied the P Cygni profiles of the new *Atlas of Ultraviolet P Cygni profiles* from Snow et al. (1994). The P Cygni profiles are all plotted in *velocity* rather than in wavelength, so that it is easy to compare the profiles of different lines and different ionization stages. We will show below that this comparison produces much more reliable estimates of v_∞ than available up to now.

We will show in §§ 3 and 4 that the observed terminal velocities of stellar winds are “high” with $v_\infty/v_{\text{esc}} \simeq 2.6$ for O and early-B stars but that this ratio drops by about a factor of 2 to $v_\infty/v_{\text{esc}} \simeq 1.3$ for stars at the cool side of $T_{\text{eff}} \simeq 21,000$ K, i.e., near spectral type B1. A similar, but smaller discontinuity might be present at lower $T_{\text{eff}} \simeq 10,000$ K but the evidence is not very strong.

In § 5 we compare the observed values of v_∞ with predictions for radiation-driven wind models, and we derive the force multiplier parameter α for all the stars by calculating radiation-driven wind models for each star with the simplified method of Kudritzki et al. (1989). We show that α is discontinuous near $T_{\text{eff}} \simeq 21,000$ K. In § 6 we discuss the nature of the bistability, and we present evidence that the mass-loss rate of supergiants is also discontinuous with a discontinuous increase on the cool side of the bistability limit. We discuss the possibility that bistability of stellar winds is responsible for the formation of disks around B[e]-supergiants. The conclusions are in § 7.

2. OBSERVATIONS

2.1. *Atlas of P Cygni Profiles*

The observations are from the *Atlas of Ultraviolet P Cygni Profiles* by Snow et al. (1994, hereafter called the Atlas). It contains the IUE spectra of 232 stars of spectral types from O3 to F8. It is the largest atlas of homogeneously reduced IUE spectra available. The computer version of this atlas, which is available through NASA's Astrophysics Data System, contains the full spectra while the printed version contains the P Cygni profiles of He II, C II, C III, C IV, N IV, N V, O IV, O V, Mg I, Mg II, Al II, Al III, Si II, Si III, Si VI, Ca II, and Fe II, all normalized to the continuum and plotted versus Doppler velocity. Since the profiles are expressed as a function of velocity, the Atlas is ideally suited for the study of the velocity of the winds and for the comparison between the profiles of different ions.

We have measured the terminal velocities of the winds of the stars in the Atlas. Stars which do not show evidence for mass loss in their UV resonance lines, or stars whose resonance lines only show violet absorption wings were omitted from this study, because their terminal velocity could not be measured. The 68 stars of types between O3 and A8 with an accurately measured value of the terminal velocity are listed in Table 1. The 49 stars for which the terminal velocity could only be measured with lower accuracy (see § 3) are listed in Appendix A.

The atlas numbers of the stars are kept for easy identification of the stars and for inspection of the profiles in the Atlas.

2.2. *Stellar Parameters*

The stellar parameters of the program stars, listed in Table 1, are from the Atlas. Since the stellar parameters are important in this study, we briefly recapitulate their sources. For O stars the data are from Garmany (1987) except for the values of T_{eff} which are from Chlebowski & Garmany (1991). This temperature scale is based on the results of several non-LTE

studies of O stars and agrees well with the temperatures derived by Herrero et al. (1992). For the other stars the data are derived from their spectral types using the relations by Schmidt-Kaler (1982). The distances are derived from cluster memberships (Humphreys 1978) or from the relation between spectral type and M_v (Schmidt-Kaler 1982). The masses are derived from T_{eff} and L using the evolutionary tracks by Maeder (1990) which include convective overshooting and mass loss. We consider these values more reliable than those derived from spectroscopic gravities because of the sensitivity of the line profiles to the details of the model atmospheres. For most stars we adopted a He/H ratio of 0.11 by number, as derived from the non-LTE studies by Herrero et al. (1992). For O stars of luminosity classes I, II, and III and for supergiants of types B0 and B1.5, we adopted a ratio of He/H = 0.25 following the results of Herrero et al. For the calculation of the radiation pressure by free electrons we assumed that H is fully ionized in the winds of O and B stars, half-ionized in the winds of A stars, and mostly neutral in the winds of F stars. Helium is assumed to be twice ionized in the winds of stars with $T_{\text{eff}} \geq 35,000$ K, 1.5 times ionized between $30,000 \leq T_{\text{eff}} < 35,000$ K, singly ionized at $25,000 \leq T_{\text{eff}} < 30,000$ K, 0.5 times ionized at $15,000 \leq T_{\text{eff}} \leq 25,000$ K, and neutral at lower temperatures (see the Atlas).

Table 1 gives the program stars, their spectral types, T_{eff} , M_{bol} , R_* , M_* , the ratio between the radiative acceleration due to electron scattering and gravity Γ_e , and the effective escape velocity, v_{esc} . For some stars the values of v_{esc} are slightly different from the ones listed in the Atlas, due to a minor change in the adopted He/H ratio. Figure 1 shows the location

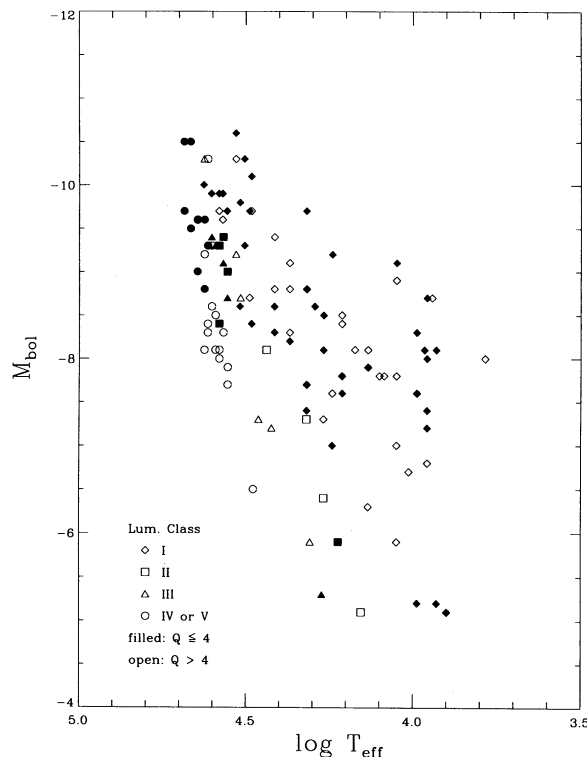


FIG. 1.—The location in the H-R diagram of the 117 stars for which v_∞ was measured. Luminosity classes are indicated by different symbols. Filled symbols refer to stars with an accurate determination of v_∞ within 10%. Open symbols refer to stars with a less accurately determined value of v_∞ .

TABLE 1
THE PRIMARY PROGRAM STARS

Atlas number	HD number	Spectral type	T_{eff} (K)	M_{bol}	R_* ($R_{\odot}$)	M_* ($M_{\odot}$)	Γ_e	v_{esc} (km s^{-1})
1	93250	O3 V((f))	48500	-10.5	15.6	82.0	0.38	1088
2	303308	O3 V((f))	48500	-9.7	10.8	60.0	0.25	1241
3	66811	O4 I(n)f	42400	-10.0	16.7	59.0	0.34	942
4	46223	O4 V((f))	46400	-9.5	11.0	52.0	0.25	1166
5	96715	O4 V((f))	46400	-10.5	17.4	81.0	0.41	1030
6	14947	O5 If+	40300	-9.9	17.3	52.0	0.33	852
8	15629	O5 V((f))	44300	-9.6	12.3	50.0	0.28	1041
10	93204	O5 V((f))	44300	-9.0	9.4	42.0	0.19	1162
11	210839	O6 I(n)f	38200	-9.9	19.6	51.0	0.35	795
13	93130	O6 III(f)	40200	-9.4	13.8	43.0	0.25	923
15	91572	O6 V((f))	42200	-8.8	9.6	38.0	0.18	1114
17	101190	O6 V((f))	42200	-9.6	13.9	48.0	0.30	964
18	163758	O6.5 Iaf	37200	-9.9	20.1	50.0	0.34	762
20	69464	O6.5 Ib(f)	37200	-9.9	20.1	49.0	0.35	750
21	190864	O6.5 III(f)	39200	-9.3	14.0	42.0	0.24	922
26	101436	O6.5 V	41200	-9.3	12.4	42.0	0.25	969
27	192639	O7 Ib(f)	36100	-9.7	19.6	43.0	0.34	725
28	34656	O7 II(f)	38100	-8.4	9.9	30.0	0.15	993
29	151515	O7 II(f)	38100	-9.3	14.9	41.0	0.25	882
32	162978	O7.5 II((f))	37100	-9.4	16.0	40.0	0.27	810
33	186980	O7.5 III((f))	37100	-9.1	13.9	35.0	0.23	834
37	151804	O8 Iaf	34000	-10.6	34.0	70.0	0.47	635
38	175754	O8 II((f))	36000	-9.0	14.2	34.0	0.22	825
39	36861	O8 III((f))	36000	-8.7	12.3	30.0	0.19	849
44	112244	O8.5 Iab(f)	33000	-9.8	25.2	46.0	0.35	674
45	74194	O8.5 Ib(f)	33000	-8.6	14.5	28.0	0.19	776
46	96917	O8.5 Ib(f)	33000	-9.8	25.2	46.0	0.35	674
48	149404	O9 Ia	32000	-10.3	33.9	60.0	0.43	625
49	210809	O9 Iab	32000	-9.3	21.4	38.0	0.27	709
56	30614	O9.5 Ia	30900	-9.7	27.6	43.0	0.34	629
57	188209	O9.5 Iab	30900	-9.7	27.6	43.0	0.34	629
58	218915	O9.5 Iab	30900	-9.7	27.6	43.0	0.34	629
64	152405	O9.7 Ib-II	30500	-8.4	15.3	25.0	0.17	716
65	152424	OC9.7 Ia	30500	-10.1	33.4	52.0	0.39	589
69	91969	B0 Ia	26000	-8.6	22.9	25.0	0.20	573
71	204172	B0 Ib	26000	-8.3	20.0	23.0	0.16	602
82	213087	B0.5 Ibe	23400	-8.2	23.4	21.0	0.17	533
87	96248	B1 Iab	20800	-8.8	38.9	25.0	0.23	430
88	152236	B1 Iape	20800	-9.7	58.9	34.0	0.39	362
90	154090	B1 Iac	20800	-8.8	38.9	26.0	0.22	441
91	109867	B1 Ia	20800	-8.8	38.9	26.0	0.22	441
92	47240	B1 Ib	20800	-7.7	23.4	17.0	0.12	488
93	52382	B1 Ib	20800	-7.4	20.4	17.0	0.09	532
95	157246	B1 Ib	20800	-7.7	23.4	17.0	0.12	488
103	106343	B1.5 Ia	19700	-8.6	40.7	24.0	0.23	426
104	148379	B1.5 Iape	19700	-8.6	40.7	24.0	0.23	426
105	190603	B1.5 Iac	19700	-8.6	40.7	24.0	0.23	426
113	41117	B2 Iave	18500	-8.5	43.4	25.0	0.19	425
115	206165	B2 Ib	18500	-8.1	35.8	19.0	0.18	410
130	92964	B2.5 Iac	17400	-9.2	68.4	29.0	0.33	339
132	116084	B2.5 Ib	17400	-7.0	24.8	15.0	0.08	464
133	166596	B2.5 III	18700	-5.3	9.8	9.7	0.03	609
140	198478	B3 Iac	16300	-7.6	36.3	17.0	0.12	394
143	43384	B3 Ib	16300	-7.8	39.8	19.0	0.13	396
144	51309	B3 II	16700	-5.9	16.3	11.0	0.04	501
150	79186	B5 Ia	13600	-7.9	62.4	18.0	0.17	312
168	34085	B8 Iac	11200	-9.1	153.3	26.0	0.32	213
186	92207	A0 Iac	9730	-8.3	143.8	22.0	0.09	232
187	223960	A0 Iac	9730	-7.6	104.2	14.0	0.08	219
188	46300	A0 Ib	9730	-5.2	34.5	9.5	0.01	323
191	12953	A1 Iac	9230	-8.1	146.4	19.0	0.09	215
192	14433	A1-3 Ia	9080	-7.4	107.6	17.0	0.05	239
193	197345	A2 Iac	9080	-8.7	195.8	24.0	0.12	203
194	14489	A2 Ia	9080	-8.0	141.9	20.0	0.08	223
195	207260	A2 Ia	9080	-7.2	98.2	16.0	0.05	243
199	17378	A5 Ia	8510	-8.1	173.1	20.0	0.09	203
200	59612	A5 Ib	8510	-5.2	45.5	9.5	0.01	283
208	85123	A8 Ib	7950	-5.1	49.8	9.4	0.01	269

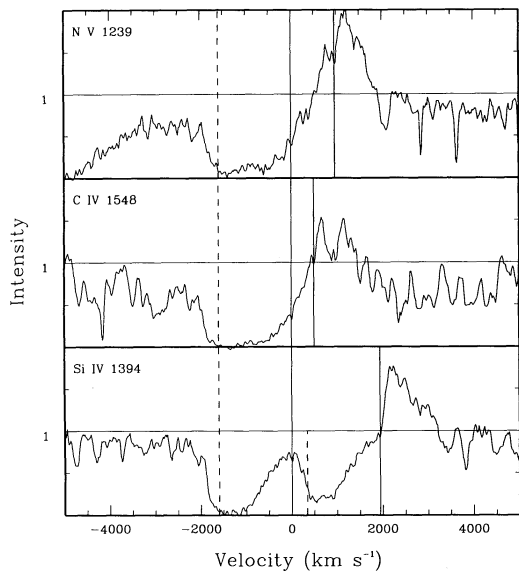


FIG. 2a

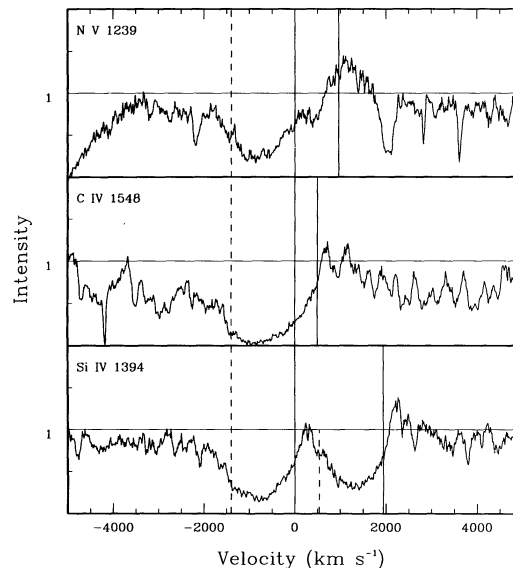


FIG. 2b

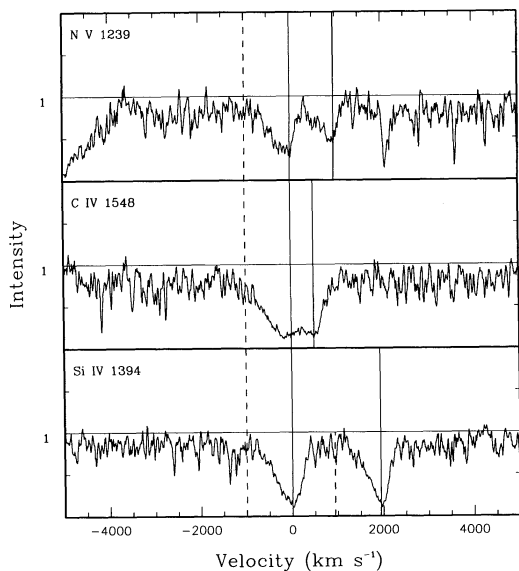


FIG. 2c

FIG. 2.—Examples of the three methods for determining v_∞ : (a) from the extent of the black absorption trough, (b) from strong but unsaturated P Cygni profiles, (c) from extended absorption wings. Full and dashed vertical lines indicate the rest wavelength of the lines and v_∞ , respectively. All stars in Table 1 have v_∞ determined from the black troughs.

in the Hertzsprung-Russell diagram of the 117 stars for which we measured v_∞ .

3. TERMINAL VELOCITIES

3.1. Methods for Measuring v_∞

The terminal velocities of the winds can be measured in different ways, depending on the types of profiles.

Traditionally, v_∞ was measured from the wavelength or velocity, v_{edge} , where the violet edge of the P Cygni profiles reach the continuum. However, Hamann (1980) and Lamers, Cerruti-Sola, & Perinotto (1987), in their introduction of the SEI method, have shown that chaotic motions in the winds

may extend v_{edge} beyond v_∞ and that v_{edge} can overestimate v_∞ considerably. Subsequently, Groenewegen, Lamers, & Pauldrach (1989, hereafter GLP) have shown from detailed fitting of the line profiles with the SEI method, that there is indeed strong evidence for large chaotic motions, probably due to shocks (Lucy 1984), in the winds of O and early-B stars. The velocities of the chaotic motions at large distances from the star, which affect the violet absorption edges of the profiles, can be as large as 0.10 to 0.20 v_∞ which corresponds to 200–400 km s^{-1} for a typical O star.

In this study we have adopted three methods for the determination of v_∞ .

1. The studies of Lamers, Cerruti-Sola, & Perinotto (1987) and of Groenewegen & Lamers (1989) have shown that v_∞ can be estimated quite accurately from the violet extent of the black absorption trough of saturated P Cygni profiles. This method was used to measure v_∞ for all the stars listed in Table 1.

2. For lines which are not strongly saturated and which do not show a black trough but which still clearly showed P Cygni profile in their UV spectrum, we derived v_∞ from the extent of the absorption, taking the possible presence of chaotic motions into account. The accuracy of this determination is less than for the stars with saturated P Cygni profiles. However, for most stars we can compare the P. Cygni profiles of different ions which results in a more reliable estimate of v_∞ .

3. If a star shows no P Cygni profiles but only extended violet absorption wings, we used the extent of the wings to find a lower limit of v_∞ .

Prinja, Barlow, & Howarth (1990, hereafter PBH) have shown that v_∞ can also be derived from the final velocity reached by the discrete absorption components (DACs) that move through the P Cygni profiles from small to large negative velocities. For the stars in common with GLP, the values of v_∞ derived from the DACs by PBH agree very well with those derived from the line fitting using the SEI method. As the Atlas contains only a snapshot IUE spectrum of each star, we cannot use this method. The DACs present in the profiles of the Atlas would only provide a lower limit to v_∞ .

Examples of the three types of measurements are shown in Figure 2. Figure 2a shows an example of a star with saturated P Cygni profiles where v_∞ can be derived from the extent of the black trough. Figure 2b shows an example of a star with clear but unsaturated P Cygni profiles. Figure 2c shows an example of a star with P Cygni profiles that have gradually rising violet absorption wings. The actual determinations of v_∞ were made by lining up the profiles of UV resonance lines and the subordinate lines with P Cygni profiles in velocity scale and measuring v_∞ simultaneously from all these profiles.

3.2. Blending Lines

The determination of v_∞ from the observed profiles can be hampered by the presence of photospheric lines near the violet edge of the P Cygni profiles. This is especially the case for lines of C IV and Si IV in the UV spectra of O and early-B stars, and in the spectra of C II lines in late-B and A-type stars. The violet side of the C IV lines contain many blends due to the photospheric metal lines, mainly Fe. This has led to severe overestimates of v_∞ by $\sim 500 \text{ km s}^{-1}$ in earlier studies.

In order to avoid the confusion by photospheric blends, we have used comparison spectra of stars in the Atlas with weak or no P Cygni profiles (usually main-sequence stars) for each spectral type. We stress that this comparison is essential to avoid overestimating v_∞ from P Cygni profiles due to photospheric blends.

3.3. Measured Terminal Velocities

The measured values of v_∞ for the program stars are listed in Table 2, together with their uncertainty. This uncertainty is estimated from a simultaneous visual inspection of all the P Cygni profiles of different ions in the UV spectrum of a star. It is *not* a standard deviation but a conservative guess of the possible range of v_∞ . Column (5) (line code) gives information about the lines of the two ions that provide the best indication for v_∞ for each star. The ions are indicated by a number and the type of profile is indicated by a letter. The codes for the ions and the types of profile from which v_∞ was determined are

listed in Table 3. We stress that v_∞ was not determined from the lines of only two ions but from the lines of all ions of the Atlas, usually six to eight lines.

The terminal velocity or its upper limit for stars with unsaturated P Cygni profiles are listed in Appendix A. They will not be used in this paper.

Each measurement of v_∞ has a quality factor, Q , in Table 2 column (6) which indicates the reliability of the v_∞ determination. This was done because the accuracy of v_∞ depends largely on the types of P Cygni profiles available. The quality factor depends on the estimated accuracy of v_∞/v_{esc} and on the method used for the determination of v_∞ . The code for Q is given in Table 4. The most reliable determinations of v_∞ , i.e., those with $Q = 1$, are based on the black trough of the saturated P Cygni profiles of at least two different ions and for each of these ions the value of v_∞ could be measured with an accuracy better than 10%.

We believe that our estimates of v_∞ are more reliable than most other determinations in the literature, except those based on detailed modeling such as the ones from GLP, because (1) we used a large set of homogeneous spectra, (2) we carefully avoided the effects of photospheric blends, and (3) we derived v_∞ from the UV lines of different ions simultaneously.

Hereafter, we will only use the most accurate measurements of v_∞ , i.e., those derived from saturated P Cygni profiles which have $Q = 1-4$. They are listed in Table 1. The terminal velocities with $Q \geq 5$ are listed in Appendix A.

3.4. Comparison with Other Determinations of v_∞

Our values of v_∞ are compared with two other sets of determinations: those from GLP derived by line fitting and those from PBH derived by several methods.

Figure 3a shows the comparison between our values and the ones of GLP for stars of types O to B1 in common in both sets of data. The agreement is excellent and well within the quoted uncertainty of both sets. Figure 3b shows the comparison of our measurements of v_∞ with PBH for stars of types O to B3 in common in both data sets. The agreement is very good but

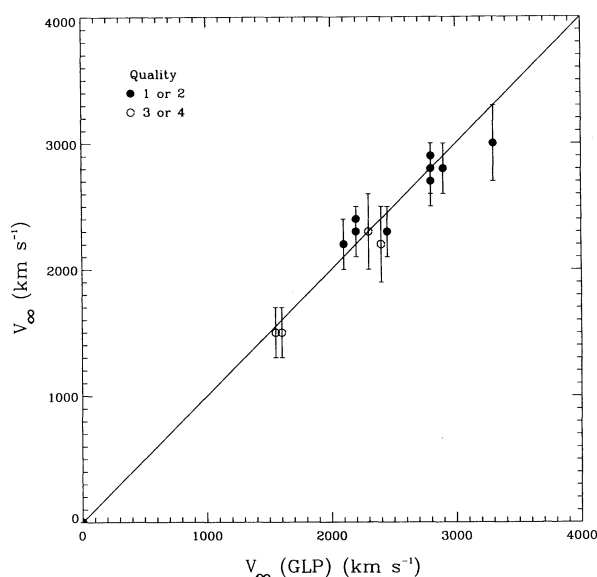


FIG. 3a

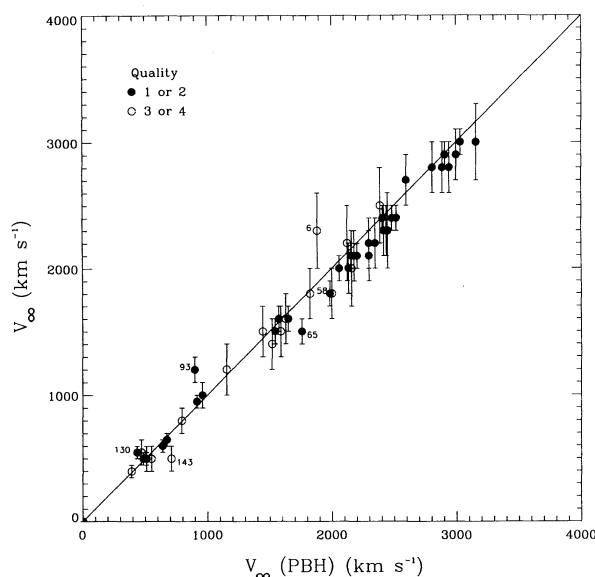


FIG. 3b

FIG. 3.—Comparison between our measurements of v_∞ and those of (a) GLP and (b) PBH. The agreement is very good and within the errors, except for some stars indicated by their Atlas number.

TABLE 2
TERMINAL VELOCITIES OF THE PRIMARY PROGRAM STARS

Atlas number	T_{eff} (K)	v_{esc} (km s^{-1})	v_{∞} (km s^{-1})	line code ^a	Q^b	v_{∞} (PBH) (km s^{-1})	v_{∞} (GLP) (km s^{-1})	$v_{\infty}/v_{\text{esc}}$	v_{∞} (pred) (km s^{-1})	α_{emp}
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
1	48500	1088	3000 ± 300	1b,2e	2	3160	3300 ± 200	2.75 ± 0.27	3800	0.536 ± 0.042
2	48500	1241	3000 ± 100	1b,2b	1	3035		2.41 ± 0.08	4333	0.488 ± 0.036
3	42400	942	2400 ± 100	1b,2b	1	2485	2200 ± 60	2.54 ± 0.10	3370	0.506 ± 0.035
4	46400	1166	2900 ± 100	2b	2	2910	2800 ± 60	2.48 ± 0.08	4103	0.498 ± 0.036
5	46400	1030	2900 ± 200	2b	2	3000		2.81 ± 0.19	3628	0.541 ± 0.034
6	40300	852	2300 ± 300	2b,3b	3	1885	2300 ± 70	2.69 ± 0.35	3077	0.528 ± 0.052
8	44300	1041	2800 ± 200	2b	2	2810	2900 ± 70	2.68 ± 0.19	3696	0.525 ± 0.036
10	44300	1162	2800 ± 200	2b	2	2890	2800 ± 60	2.40 ± 0.17	4127	0.487 ± 0.044
11	38200	795	2200 ± 200	2b,3b	1	2300	2100 ± 60	2.76 ± 0.25	2882	0.535 ± 0.040
13	40200	923	2500 ± 300	2b	4	2390		2.70 ± 0.32	3337	0.530 ± 0.048
15	42200	1114	2400 ± 100	1b,2b	1	2410		2.15 ± 0.08	3989	0.441 ± 0.050
17	42200	964	2800 ± 200	2b	2	2945	2900 ± 150	2.90 ± 0.20	3452	0.551 ± 0.033
18	37200	762	2300 ± 200	2b,3b	1	2420	2200 ± 70	3.01 ± 0.26	2765	0.561 ± 0.035
20	37200	750	2100 ± 200	1b,3e	2	2300		2.80 ± 0.26	2721	0.540 ± 0.040
21	39200	922	2300 ± 200	1b,2b	1	2440	2450 ± 150	2.49 ± 0.21	3338	0.499 ± 0.046
26	41200	969	2700 ± 200	1e,2b	2	2600	2800 ± 150	2.78 ± 0.20	3487	0.539 ± 0.036
27	36100	725	2100 ± 200	1b,2b	1	2180		2.89 ± 0.27	2621	0.550 ± 0.038
28	38100	993	2100 ± 100	1b,2b	1	2155		2.11 ± 0.10	3598	0.433 ± 0.053
29	38100	882	2400 ± 100	1b,2b	1	2520		2.72 ± 0.11	3196	0.531 ± 0.030
32	37100	810	2200 ± 200	1b,2b	1	2350		2.71 ± 0.24	2939	0.530 ± 0.041
33	37100	834	2100 ± 100	1b,2b	1	2205		2.51 ± 0.11	3028	0.501 ± 0.036
37	34000	635	1500 ± 200	1b,2b	3	1445	1600 ± 70	2.36 ± 0.31	2226	0.480 ± 0.059
38	36000	825	2000 ± 100	1b,3b	1	2060		2.42 ± 0.12	2998	0.488 ± 0.039
39	36000	849	2200 ± 300	1d,2b	4	2125	2400 ± 150	2.59 ± 0.35	3084	0.512 ± 0.058
44	33000	674	1600 ± 100	1b,3b	1	1575		2.37 ± 0.14	2430	0.480 ± 0.043
45	33000	776	2000 ± 300	1b,3b	3	2160		2.57 ± 0.38	2829	0.510 ± 0.063
46	33000	674	1800 ± 200	1b,3b	3	2000		2.67 ± 0.29	2430	0.524 ± 0.048
48	32000	625	2300 ± 300	2b,3b	3	2450		3.68 ± 0.48	2205	0.634 ± 0.049
49	32000	709	2000 ± 200	1b,3b	1	2135		2.82 ± 0.28	2588	0.543 ± 0.041
56	30900	629	1500 ± 200	1b,3b	3	1590	1550 ± 60	2.38 ± 0.31	2241	0.482 ± 0.060
57	30900	629	1600 ± 100	1b,3b	1	1650		2.54 ± 0.15	2241	0.505 ± 0.038
58	30900	629	1800 ± 100	1b,2b	1	1985		2.86 ± 0.15	2241	0.546 ± 0.030
64	30500	716	1800 ± 200	2b,3e	4	1825		2.51 ± 0.27	2616	0.501 ± 0.051
65	30500	589	1500 ± 100	2b,3b	1	1760		2.54 ± 0.16	2056	0.508 ± 0.038
69	26000	573	1500 ± 100	2b,3b	1	1545		2.61 ± 0.17	1816	0.515 ± 0.037
71	26000	602	1600 ± 200	2b,3e	4	1630		2.65 ± 0.33	1911	0.522 ± 0.051
82	23400	533	1400 ± 200	2b,3b	3	1520		2.62 ± 0.37	1533	0.516 ± 0.057
87	20800	430	650 ± 50	2b,3b	1	675		1.51 ± 0.11	1116	0.317 ± 0.059
88	20800	362	400 ± 50	3b,5b	3	390		1.10 ± 0.13	919	0.205 ± 0.050
90	20800	441	950 ± 50	2b,3b	1	915		2.15 ± 0.11	1144	0.456 ± 0.039
91	20800	441	1200 ± 200	2b,3b	3	1155		2.72 ± 0.45	1144	0.543 ± 0.065
92	20800	488	1000 ± 100	2b,3b	1	960		2.04 ± 0.20	1267	0.426 ± 0.056
93	20800	532	1200 ± 100	2b,3e	2	900		2.25 ± 0.18	1381	0.461 ± 0.052
95	20800	488	900 ± 200	2b,3e	4			1.84 ± 0.40	1267	0.386 ± 0.101
103	19700	426	800 ± 100	2b,3b	3	795		1.87 ± 0.23	1074	0.402 ± 0.064
104	19700	426	500 ± 100	2b,3b	3	510		1.17 ± 0.23	1074	0.211 ± 0.082
105	19700	426	500 ± 50	2b,3b	1	485		1.17 ± 0.11	1074	0.211 ± 0.043
113	18500	425	500 ± 50	3b,10b	1	510		1.17 ± 0.11	1058	0.210 ± 0.043
115	18500	410	600 ± 50	2b,3b	1	640		1.46 ± 0.12	1011	0.302 ± 0.062
130	17400	339	550 ± 50	2b,5b	1	435		1.62 ± 0.14	807	0.361 ± 0.053
132	17400	464	500 ± 100	2b,5b	3	550		1.07 ± 0.21	1201	0.167 ± 0.079
133	18700	609	700 ± 50	2d,6b	2			1.14 ± 0.08	1551	0.178 ± 0.030
140	16300	394	550 ± 100	3b,5b	3	470		1.39 ± 0.25	1009	0.281 ± 0.094
143	16300	396	500 ± 100	2b,3b	3	710		1.26 ± 0.25	1015	0.239 ± 0.107
144	16700	501	700 ± 100	2b,3e	4			1.39 ± 0.19	1309	0.264 ± 0.092
150	13600	312	450 ± 50	2b,5b	3			1.44 ± 0.16	772	0.306 ± 0.066
168	11200	213	350 ± 50	5b,10b	3			1.64 ± 0.23	511	0.393 ± 0.067
186	9730	232	200 ± 30	9b	4			0.86 ± 0.12	600	0.101 ± 0.053
187	9730	219	120 ± 20	9b	4			0.54 ± 0.09	561	
188	9730	323	180 ± 30	9b	4			0.55 ± 0.09	834	
191	9230	215	170 ± 20	9b	4			0.79 ± 0.09	537	
192	9080	239	150 ± 20	9b	4			0.62 ± 0.08	610	
193	9080	203	200 ± 30	5b,9b	3			0.98 ± 0.14	498	0.160 ± 0.059
194	9080	223	150 ± 20	5b,9b	3			0.67 ± 0.08	563	
195	9080	243	150 ± 20	8b,9b	3			0.61 ± 0.08	622	
199	8510	203	180 ± 30	9b	4			0.88 ± 0.14	506	0.117 ± 0.059
200	8510	283	180 ± 20	9b	4			0.63 ± 0.07	715	
208	7950	269	200 ± 30	9b,13d	4			0.74 ± 0.11	674	

^a See Table 3.

^b Quality code as defined in Table 4.

TABLE 3
CODES FOR DETERMINATION OF v_{∞}
A. LINES

Number	Ion	Number	Ion
1	N v $\lambda 1239$	8	Si II $\lambda 1527$
2	C IV $\lambda 1548$	9	Mg II $\lambda 2796$
3	Si IV $\lambda 1394$	10	Al III $\lambda 1855$
4	C III $\lambda 1176$	11	Al II $\lambda 1671$
5	C II $\lambda 1335$	12	Ca II $\lambda 1650$
6	Si III $\lambda 1207$	13	Fe II $\lambda 2586$
7	Si III $\lambda 1895$	14	Al II $\lambda 2669$

B. TYPES

Codes for Ions	Types of Profiles
b	Extent of black trough
e	Extent of absorption part of unsaturated P Cygni profile
d	Discrete absorption component
w	Violet absorption wing

there are some exceptions. These are indicated by the number of the star in the Atlas. Since PBH do not give the uncertainty of their measurements, we cannot judge the significance of these differences.

4. RATIO $v_{\infty}/v_{\text{esc}}$

The theory of radiation-driven stellar winds predicts that the terminal velocity of the winds scales with the escape velocity approximately as $v_{\infty}/v_{\text{esc}} \sim [\alpha/(1-\alpha)]^{1/2}$, so the proportionality factor depends on α which depends on the effective temperature of the star (CAK; Abbott 1982; Pauldrach et al. 1986). The empirical values of $v_{\infty}/v_{\text{esc}}$ are listed in Table 2, column (9), together with their uncertainties derived from the value of $\Delta(v_{\infty})$. We have plotted the ratio $v_{\infty}/v_{\text{esc}}$ as a function of T_{eff} in Figure 4a. In order to avoid confusion where points in this figure would be on top of one another, we have applied a small horizontal shift of $\Delta \log(T_{\text{eff}}) = \pm 0.01$ near $T_{\text{eff}} \simeq 21,000$ K.

The figure shows that $v_{\infty}/v_{\text{esc}} \simeq 2.51 \pm 0.27$ for stars with $\log(T_{\text{eff}}) > 4.40$, in good agreement with the value of 2.78 ± 0.36 found by GLP. For cooler stars the ratio $v_{\infty}/v_{\text{esc}}$ decreases to lower temperatures, reaching a value of 0.68 ± 0.14 at $3.9 < T_{\text{eff}} < 4.0$. The region in between shows a curious trend: the high value of $v_{\infty}/v_{\text{esc}} \simeq 2.5$ seems to extend all the way down from $\log(T_{\text{eff}}) = 4.7$ to below 4.4, where it drops quite steeply to $v_{\infty}/v_{\text{esc}} \simeq 1.3$ at $\log(T_{\text{eff}}) \simeq 4.32$ or $T_{\text{eff}} \simeq 21,000$ K at spectral type B1. This is shown in Figure 4b, where the data compared with a linear least-squares fit and a discontinuous fit.

TABLE 4
QUALITY CODE (Q) FOR THE DETERMINATION OF v_{∞}

Q	$\Delta(V_{\infty})/V_{\infty}$	Method ^a
1	≤ 0.10	At least two ions with method b
2	≤ 0.10	One ion with method b
3	≤ 0.20	At least two ions with method b
4	≤ 0.20	One ion with method b
5	≤ 0.30	At least one ion with method b
6	≤ 0.30	At least one ion with method b
7	...	Method e
8	...	Method w
9	...	Lower limit

^a See Table 3.

The discontinuous fit matches the data more closely. Stars near the discontinuity are indicated by their Atlas number and will be discussed below.

The ratio $v_{\infty}/v_{\text{esc}}$ is not necessarily multivalued at the temperature of $T_{\text{eff}} \simeq 21,000$ K. It is possible that the B1-type stars with small values of $v_{\infty}/v_{\text{esc}}$ have a slightly smaller value of T_{eff} than the B1 stars with a large value of $v_{\infty}/v_{\text{esc}}$. However, since the stars are all of the same spectral type, their values of T_{eff} must be very similar. So if $v_{\infty}/v_{\text{esc}}$ is not multivalued at type B1, it is at least a *very steep* function of T_{eff} near 21,000 K.

Another drop might be present near $\log(T_{\text{eff}}) \simeq 4.05$ but the evidence for this depends only on one star (number 168) at $\log(T_{\text{eff}}) = 4.049$. We will ignore this possible second jump and concentrate on the one near $T_{\text{eff}} = 21,000$ K. We will refer to this jump in $v_{\infty}/v_{\text{esc}}$ near $\log(T_{\text{eff}}) \simeq 4.32$ as the *bistability jump*.

The reality of the bistability jump is demonstrated clearly in Figure 5, where we show the P Cygni profiles of the stars in the temperature range of $20,800 < T_{\text{eff}} < 26,000$ K. They are indicated by their Atlas number in Figure 4. The stars are ordered from high to low v_{∞} . For each star we show the lines from the three ions that most clearly indicate the value of v_{∞} . These are the lines of C IV and Si IV and the lines of N v for the stars with high values of v_{∞} , C III for the stars with intermediate values of v_{∞} , and C II for the stars with low values of v_{∞} . Stars 91, 93, 92, 90, 95, 87, and 88 all have the same value of T_{eff} yet the profiles clearly show a large range in v_{∞} from 1200 to 400 km s⁻¹, and $v_{\infty}/v_{\text{esc}}$ has a range from 2.70 to 1.09. We will return to the interpretation of the bistability jump in § 6.

5. COMPARISON WITH RADIATION-DRIVEN WIND MODELS

The observed dependence of $v_{\infty}/v_{\text{esc}}$ on T_{eff} can be checked against predictions of radiation-driven wind models. The purpose of this comparison is two-fold: (1) the observed ratio $v_{\infty}/v_{\text{esc}}$ can be used to improve the radiation-driven wind models (in particular, it may lead to improved empirical estimates of the force multiplier parameter α); and (2) we want to check whether the observed bistability agrees with the predictions for radiation-driven wind models.

5.1. Method for Predicting v_{∞} for Radiation-driven Wind Models

Kudritzki et al. (1989) have derived an approximate analytical solution to the hydrodynamical equations of the radiation-driven wind models, taking into account the finite size of the star. The solution depends on the stellar parameters M_* , L_* , and R_* ; on the H/He ratio; on the ionization of H and He in the wind; and on the three force multiplier parameters α , δ , and k . The terminal velocity does not depend on k but only on α and δ . The equations for predicting v_{∞} for this analytical solution are given in Appendix B. This method predicts the terminal velocities with an accuracy better than 5% compared to the full numerical solution of the equations for radiation-driven wind models.

The predictions with this method confirm the original conclusion of the first radiation-driven wind models for a point-source star by CAK that $v_{\infty}/v_{\text{esc}} \sim [\alpha/(1-\alpha)]^{1/2}$.

5.2. Predicted Values of v_{∞}

We use the method of Kudritzki et al. (1989) to predict v_{∞} from the radiation-driven wind theory for each of our program stars. The force multiplier parameters from Abbott (1982) were adopted. These were calculated for a large range of tem-

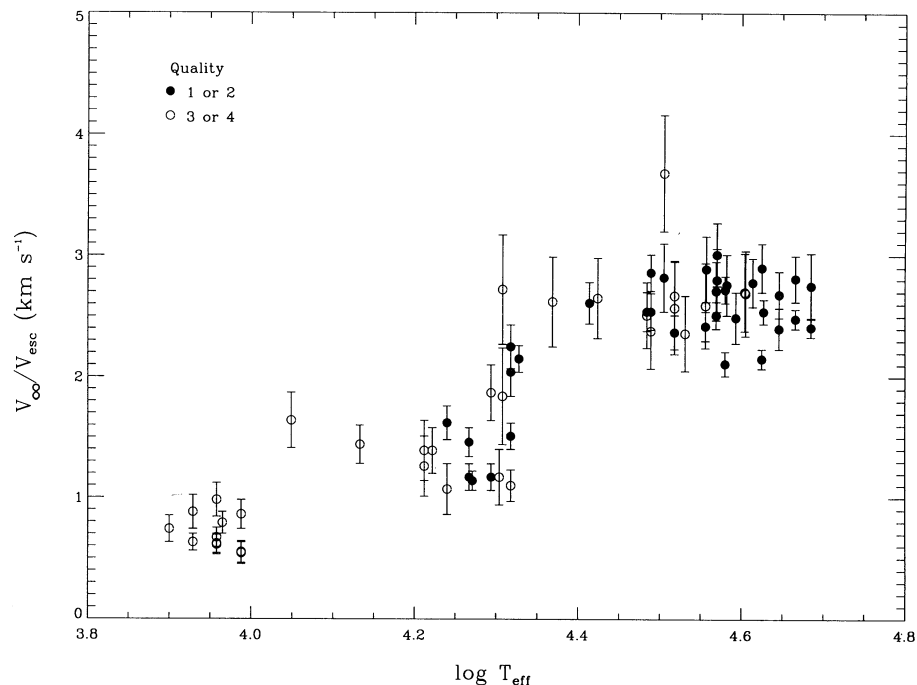


FIG. 4a

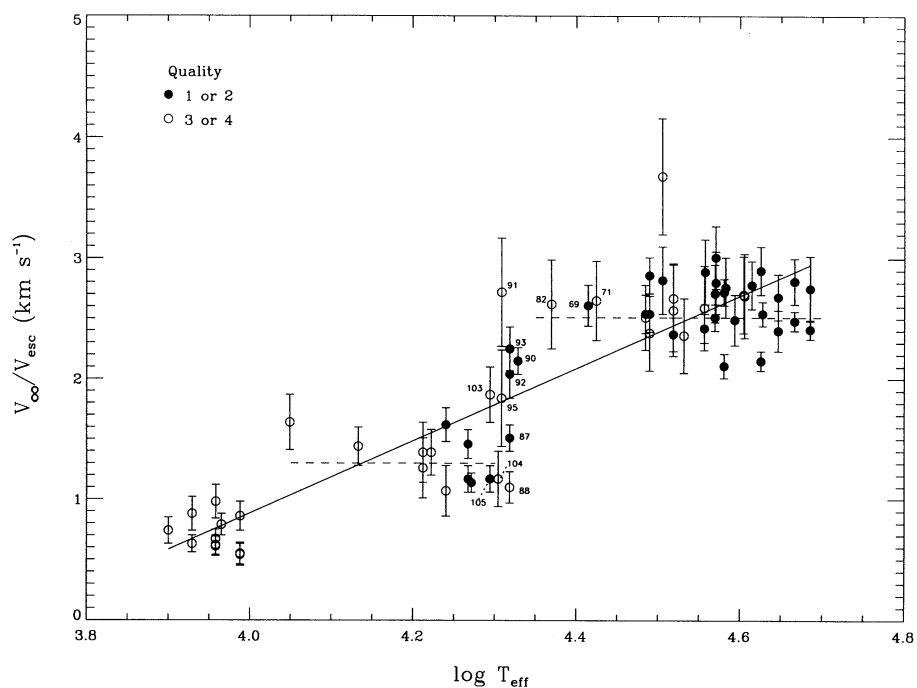


FIG. 4b

FIG. 4.—The ratio $v_{\infty}/v_{\text{esc}}$ as a function of T_{eff} . The data in (a) show that $v_{\infty}/v_{\text{esc}}$ has a high value for $\log(T_{\text{eff}}) \gtrsim 4.40$ and a low value for $\log(T_{\text{eff}}) \lesssim 4.30$ with a jump near $\log(T_{\text{eff}}) \simeq 4.32$ at spectral type B1. This is the *bistability jump*. (b) A linear least-square fit (solid line) and a step function (dashed line). The step function with $v_{\infty}/v_{\text{esc}} = 2.6$ at $\log T_{\text{eff}} \gtrsim 4.40$ and 1.3 at $\log T_{\text{eff}} \lesssim 4.30$ fits the data much better. The stars near the bistability jump are indicated by their number.

peratures from $T_{\text{eff}} = 6000$ – $50,000$ K and are based on the calculation of the contributions to the radiation pressure of more than 10^5 lines of many ionization stages for the elements from H to Zn. Abbott's values of α and δ are listed in Table 5. We adopted his values of α for supergiants since most of the stars used in this study of v_{∞} are supergiants.

Using the values of α and δ from Table 5, we calculated the predicted values of v_{∞} . The result is listed in Table 2, column (10). The predicted values of v_{∞} are compared with the observed ones in Figure 6, which shows the ratio $v_{\infty}(\text{obs})/v_{\infty}(\text{pred})$ as a function of $\log(T_{\text{eff}})$. This figure shows that for almost all stars the observed value of v_{∞} is smaller than

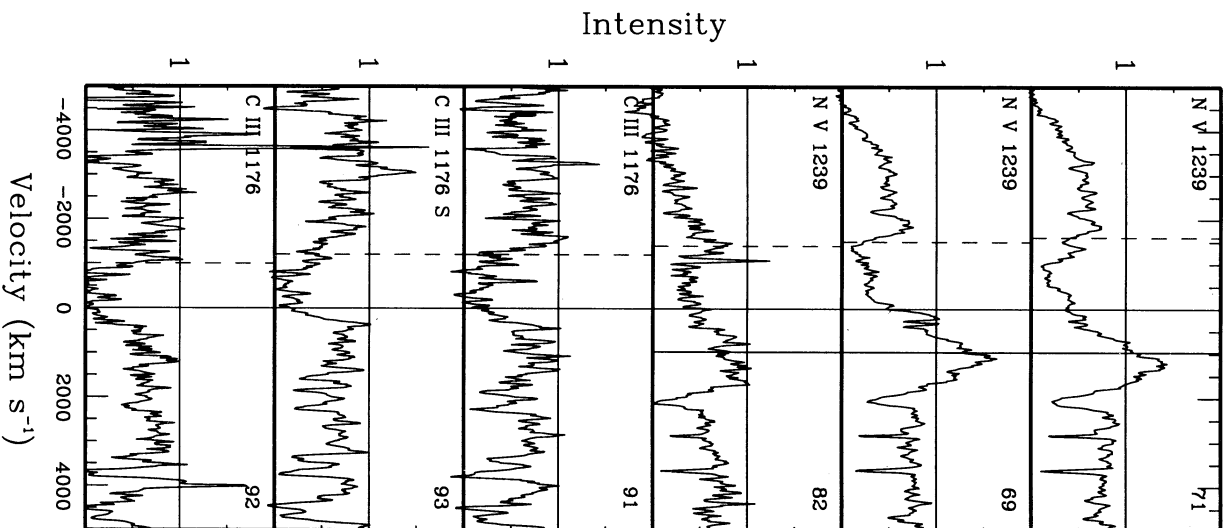
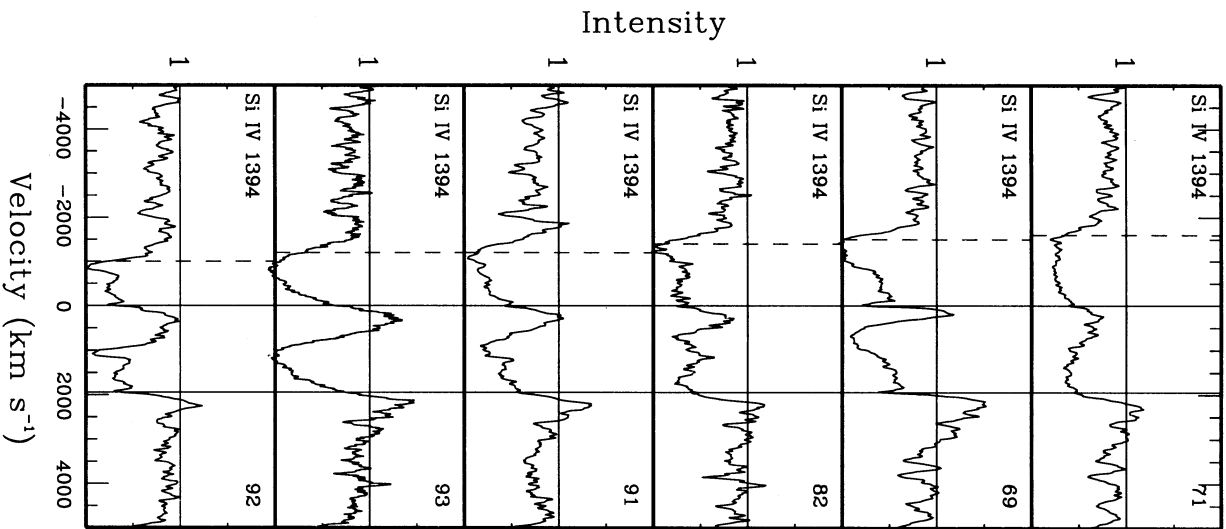
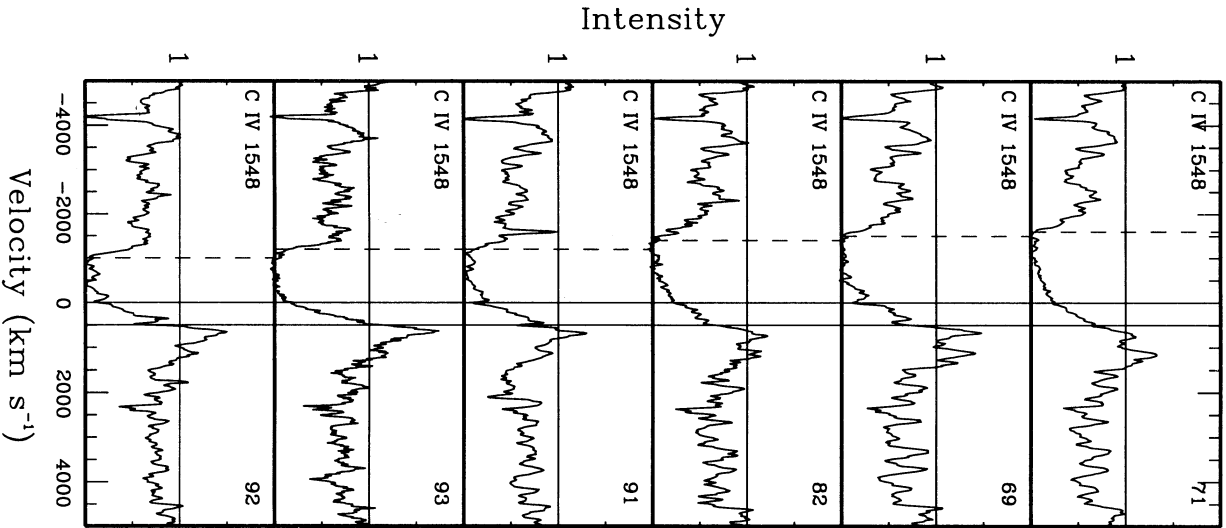


FIG. 5.—P Cygni profiles of stars near the bistability jump in order of decreasing v_{∞} . Lines from the three ions with the best indicators of v_{∞} are shown. For stars with low values of v_{∞} the C II lines are

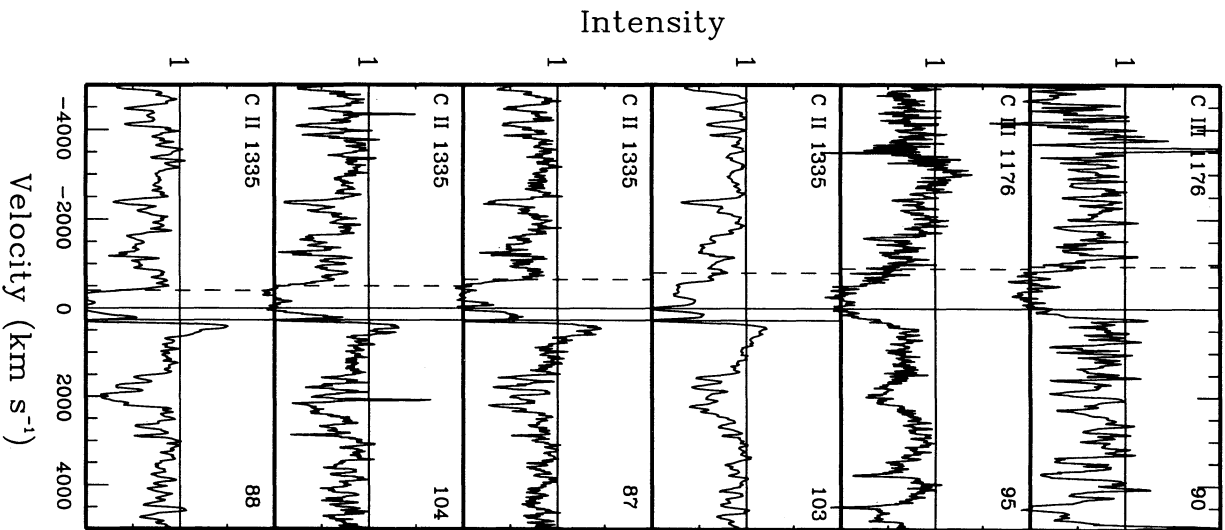
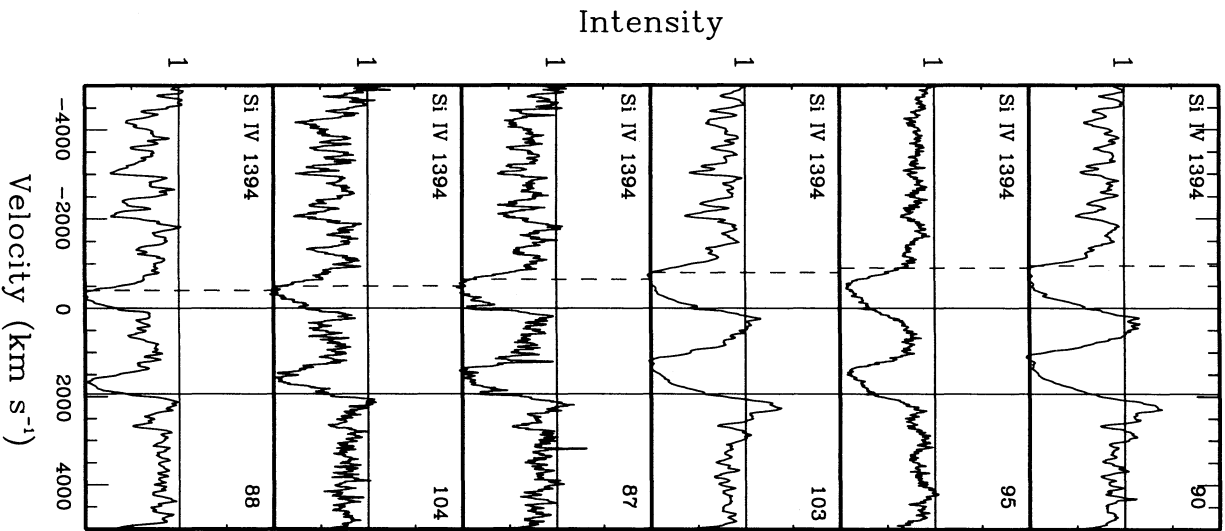
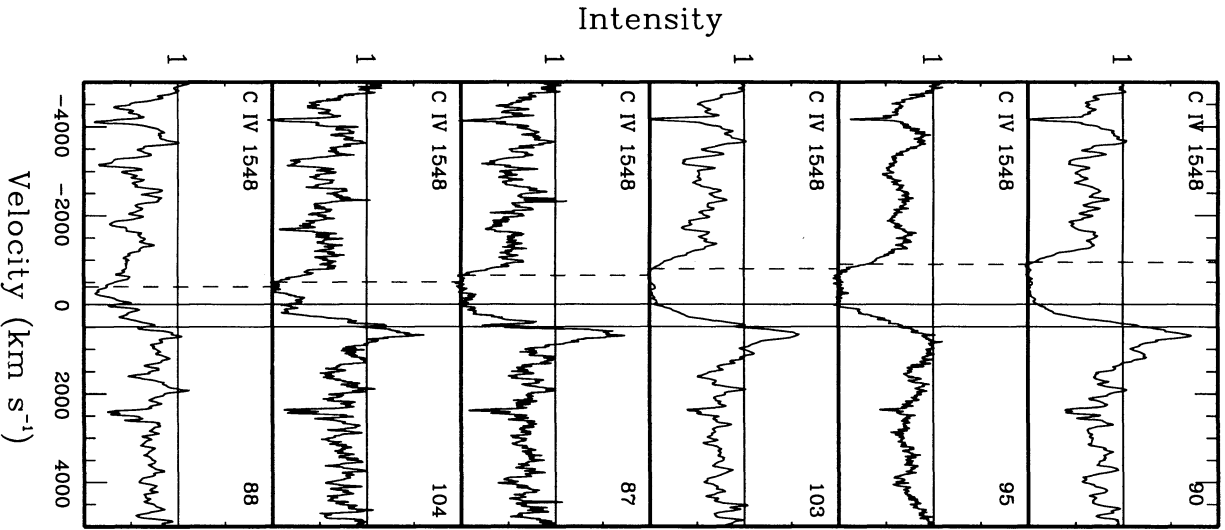


Fig. 5—Continued

TABLE 5
FORCE MULTIPLIER
PARAMETERS

T_{eff}	α	δ
6000.....	0.465	0.16
8000.....	0.542	0.02
10000.....	0.538	0.05
15000.....	0.511	0.12
20000.....	0.506	0.09
30000.....	0.609	0.12
40000.....	0.606	0.12
50000.....	0.606	0.09

the predicted value. The discrepancy runs from about a factor of 0.75 for O-type stars, about a factor 0.55 for B-type stars later than B1, to about a factor 0.30 for A-type stars. Since $v_{\infty}/v_{\text{esc}} \sim [\alpha/(1-\alpha)]^{1/2}$ this implies that α is overestimated by the models. A similar conclusion was reached by GLP when they compared the observed values of v_{∞} for O stars with predictions using the non-LTE values of α and δ by Pauldrach et al. (1990) using the masses from the evolutionary tracks. If spectroscopic masses had been adopted the discrepancy between the predicted and observed values of v_{∞} would disappear, but the location of the stars in the H-R diagram would disagree with the evolutionary models.

5.3. Empirical Determination of Force Multiplier Parameters

Using the observed values of v_{∞} we can derive empirical values of the force multiplier parameter α_{emp} for all our program stars. This is done by calculating the predicted values of v_{∞} for different values of α and iterating the method until $v_{\infty}(\text{obs}) = v_{\infty}(\text{pred})$. The resulting empirical values, α_{emp} , depend only weakly on the adopted value of δ . Therefore, we have calculated α_{emp} for three values of $\delta = 0.05, 0.10$, and 0.15

which is the range of δ 's predicted by Abbott (1982) (see Table 5). The uncertainty in α_{emp} is the result of the uncertainty in δ and in v_{∞} . Both facts were taken into account in calculating $\Delta(\alpha_{\text{emp}})$.

The resulting values of α_{emp} are listed in Table 2, column (11), and plotted versus $\log(T_{\text{eff}})$ in Figure 7. The dashed line gives the values predicted by Abbott (1982). Notice that α_{emp} is smaller than α_{pred} for almost all stars. This reflects the result that $v_{\infty}(\text{obs}) < v_{\infty}(\text{pred})$ which we found earlier.

The data in Figure 7 can be approximated by two linear relations

$$\alpha_{\text{emp}} = 0.52 \quad \text{for } 4.35 \lesssim \log(T_{\text{eff}}) \leq 4.70, \quad (2)$$

$$\alpha_{\text{emp}} = -0.71 + 0.22 \log(T_{\text{eff}}) \quad \text{for } 3.90 \lesssim \log(T_{\text{eff}}) \lesssim 4.30. \quad (3)$$

In the range of $4.30 \lesssim \log(T_{\text{eff}}) \lesssim 4.35$ the value of α_{emp} is either a very steep function of $\log(T_{\text{eff}})$ or depends on some other effects as well. (This will be discussed in the next section.)

We stress that the values of α_{emp} derived here are not necessarily identical to the values predicted by the detailed radiation-driven wind models. They are only meant to provide a simple way for predicting the value of v_{∞} for early type stars. With our values of α_{emp} , together with the adopted value of $\delta = 0.10$ and the method described in Appendix B, the terminal velocities of stars with radiation-driven winds can be predicted with an accuracy of $\sim 5\%$ if the stellar parameters are accurately known.

6. BISTABILITY JUMP NEAR $T_{\text{eff}} \approx 21,000$ K

6.1. Bistability and Ionization in the Wind

The bistability jump is related to the degree of ionization in the wind. This was explained clearly by Pauldrach & Puls

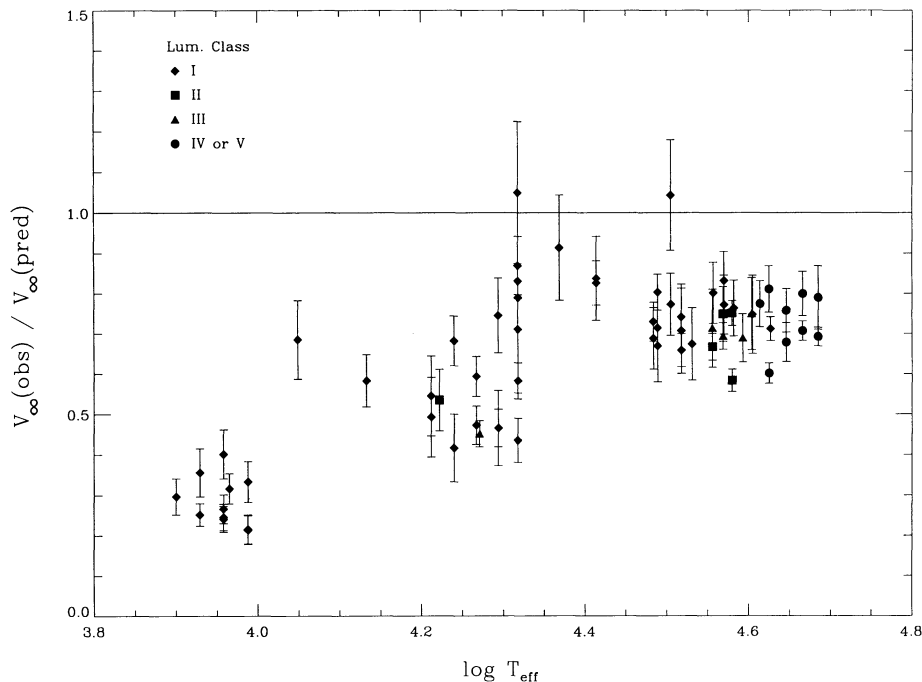


FIG. 6.—Comparison between predicted and observed values of v_{∞} as a function of T_{eff} . The observed values are systematically lower than predicted by the radiation-driven wind models with the force multipliers by Abbott (1982).

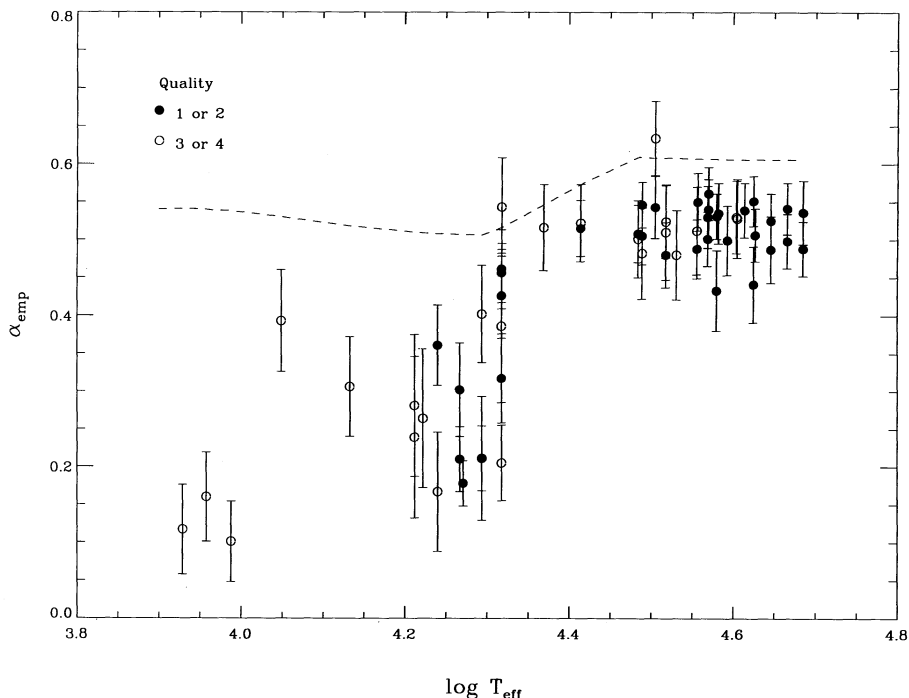


FIG. 7.—The empirical values of the force multiplier parameter α as a function of T_{eff} . The dashed line shows the predicted values. These values can be used to predict v_{∞} for any early type star with a radiation-driven wind.

(1990) in the paper where they described the discovery of the bistability for the star P Cygni. Winds which are optically thin in the Lyman continuum have a relatively “high” degree of ionization, so that the radiation pressure is mainly produced by lines in the Lyman continuum. This produces a “high” value of the force multiplier parameter α and a high terminal velocity of $v_{\infty}/v_{\text{esc}} \simeq 2.6$. Winds which are optically thick in the Lyman continuum have a relatively low degree of ionization and the radiation pressure is mainly produced by a very large number of weak metal lines in the Balmer continuum. This results in a lower value of α and a lower terminal velocity (see also Lamers & Pauldrach 1991).

Pauldrach and Puls have calculated models for a star of $T_{\text{eff}} = 19,300$ K (the temperature of P Cygni) and they varied the radius of the star and hence also the value of the effective gravity and the mass-loss rate. They found that the jump from the high to low values of v_{∞} is very steep, because the degree of ionization is very sensitive to the optical depth in the Lyman continuum: a very small change in either g_{eff} or T_{eff} near the bistability limit results in a drastic change in the mass-loss rate by $\Delta \log(\dot{M}) \simeq 0.4$ and in the terminal velocity by $\Delta \log(v_{\infty}) \simeq 0.4$ with the high value of v_{∞} corresponding to the low value of $\dot{M}$.

Abbott (1982) has calculated which elements produce most of the radiative force. From his Figures 5 and 7 one can see that near $\log(T_{\text{eff}}) \simeq 4.30$ the dominant contribution changes drastically from the lines of the C, N, and O ions at $\log(T_{\text{eff}}) > 4.30$ to the lines of ions from the Fe-group elements at $\log(T_{\text{eff}}) < 4.30$. Since the degree of ionization depends not only on temperature but also on the density, it is easy to understand why the differences between the two bistable states of the wind result in a flip-flop situation: if the wind starts to recombine, the radiative acceleration changes from high ionization, resulting in a high-velocity wind, to lower ionization which produces a lower wind velocity. But a decrease in wind velocity

in turn results in an increase in wind density, since $\rho \sim v^{-1}$, and so the wind recombines even more. This process continues until the new wind model is reached in which the wind is recombined and the radiation pressure due to the low-ionization lines produces a slow and dense wind.

We want to investigate whether the bistability jump found in this paper for normal stars near $T_{\text{eff}} \simeq 21,000$ K is also related to the degree of ionization in the wind. For this purpose we look for the presence of a correlation between the degree of ionization in the wind and the value of $v_{\infty}/v_{\text{esc}}$ or v_{∞} for the B1 stars at the jump. Figure 8 shows the profiles of the UV lines of C IV $\lambda\lambda 1548$, and 1550, C III $\lambda 1176$, and C II $\lambda\lambda 1335$, 1336 for stars with $Q = 1-3$ at the bistability limit of $T_{\text{eff}} \simeq 21,000$ K. The stars are ordered from high v_{∞} to low v_{∞} . The upper three stars with $v_{\infty} \geq 1000$ km s $^{-1}$ show strong wind lines of C IV and C III but no or only weak wind lines of C II. On the other hand, the lower three stars with $v_{\infty} \leq 950$ km s $^{-1}$ show clear C II wind lines. The lowest star (No. 88) with $v_{\infty} = 400$ km s $^{-1}$ shows a weak C IV profile and a strong P Cygni profile in C II.

There is obviously a strong correlation between the degree of ionization in the wind and the terminal velocity for stars near the bistability limit, in the sense that the high-ionization corresponds to high v_{∞} and low-ionization corresponds to low v_{∞} . This demonstrates that the observed bistability jump found for normal stars near $T_{\text{eff}} \simeq 21,000$ K in this paper is due to the same mechanism as one found in the model of P Cygni by Pauldrach & Puls (1990), viz. a drastic change in the ionization of the wind for supergiants of spectral type near B1.

6.2. Effect of Bistability on Mass-Loss Rates

We have discussed the effect of the bistability jump on v_{∞} above. Here we discuss the possible effect of the bistability on the mass-loss rates.

The stellar wind models of Pauldrach & Puls (1990) concerning the bistability of the wind of P Cygni predict that at the

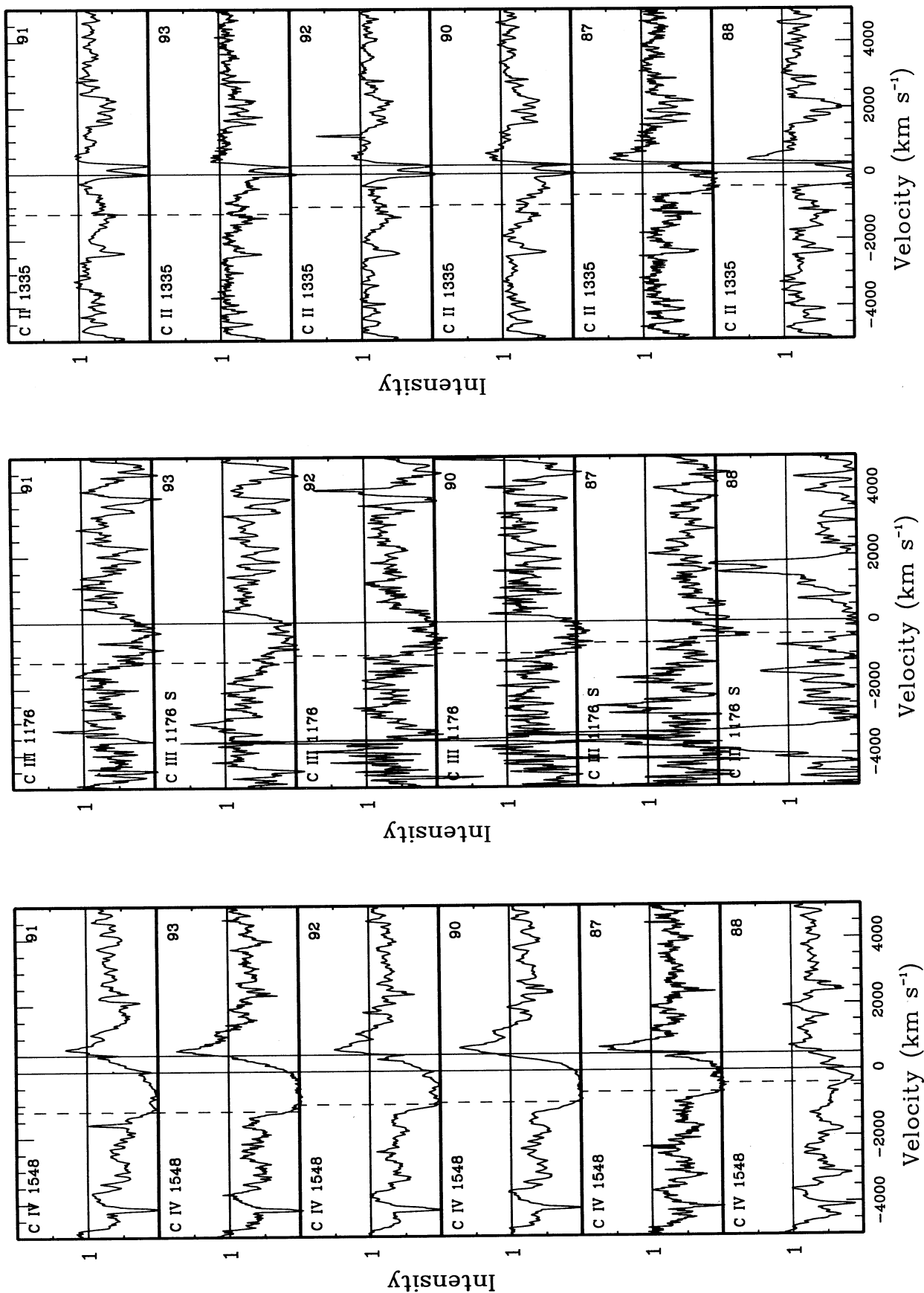


FIG. 8.—The relation between v_∞ and the ionization in the wind. The figure shows the profiles of the lines of C IV (1550 Å), C III (1176 Å) and C II (1335 Å) for stars of type B1 near the bistability jump. Notice that stars with high values of v_∞ have strong C IV lines but no C III lines (apart from interstellar components), whereas the stars with the lowest values of v_∞ have strong C II P Cygni profiles and weaker C IV lines. This demonstrates that the bistability jump is due to a change in the ionization in the wind.

bistability jump, the mass-loss rate and the terminal velocity are anticorrelated in the sense that the product $\dot{M}v_\infty$ is about constant at both sides of the jump (see Lamers & Pauldrach 1991). This implies that about the same fraction of the momentum of the radiation, L/c , is transferred into the momentum of the wind, $\dot{M}v_\infty$, independent of the degree of ionization.

Therefore, we can expect the same to occur in the normal stars near the observed bistability jump: the mass loss rate will make a jump by about a factor of 2 when crossing the bistability jump from the high-temperature side to the low-temperature side of $T_{\text{eff}} \simeq 21,000$ K.

It is difficult to test this observationally. There are as yet no reliable mass-loss determinations from UV lines for stars later than B0. The spectra in the Atlas do not give information on this. We clearly see a trend in the strength of almost all lines with v_∞ for stars near the bistability jump; however this may reflect mainly changes in the degree of ionization rather than in the mass-loss rates. We intend to make a detailed study of the mass-loss rates from the UV lines of stars near the bistability limit in the near future.

6.3. Bistability and Formation of Disks around B[e]-Stars

Lamers & Pauldrach (1991) have shown that the bistability may produce outflowing disks around rapidly rotating early-B stars. Rapidly rotating stars have higher photospheric temperatures near the poles than near the equator. Moreover, the effective gravity will be greater near the pole than at the equator. This means that a rapidly rotating star of $T_{\text{eff}} \simeq 21,000$ K can have a wind which is optically thick in the Lyman continuum near the equator (due to the smaller ionizing UV flux, the higher mass-loss rate, and the lower wind velocity) and an optically thin wind near the poles (due to the higher ionizing flux, a lower mass-loss rate, and a higher v_∞). The bistability will produce a drastic change in the wind-property at some stellar latitude between the pole and the equator: the mass flux at the equator will be about a factor 2.5 higher than at the poles and the terminal wind velocity will be about a factor 2.5 smaller. This produces a density contrast ($\rho \sim \dot{M}v_\infty$) of ~ 1 order of magnitude between the equator and the poles, with a rather steep jump at the latitude where the Lyman continuum in the wind is of order unity. The result is an equatorial outflowing disk.

Lamers & Pauldrach estimated this effect quantitatively for classical Be stars, i.e., rapidly rotating B-stars near the main sequence. They found that for these stars the mechanism by itself is not sufficient to explain the presence of the disks around Be-stars because the mass-loss rates of these stars are too small to make the wind optically thick in the Lyman continuum.

A similar estimate of the effect of the bistability of B[e]-supergiants by Lamers & Israelian (1995), shows that bistability can explain the disks around these stars, which have much higher mass-loss rates. The compression of the disk by the wind material that flows into the equatorial disk, as suggested in the wind compressed disk model of Bjorkman & Cassinelli (1993), will enhance the density contrast between the equatorial disk and the polar wind even further.

7. SUMMARY, DISCUSSION, AND CONCLUSIONS

We have carefully measured the terminal velocities of stellar winds from stars of types O3 to F8 from the UV spectra in the Atlas of Ultraviolet P Cygni profiles (Snow et al. 1994), taking

into account the possible contributions by photospheric blends to the P Cygni profiles. Our determinations agree very well with those GLP for O stars and with those of PBH for stars of types O to B3.

PBH have studied the relation between v_∞ and T_{eff} or v_{esc} . They found that v_∞ correlates with v_{esc} (with a correlation coefficient $r = 0.76$) but that v_∞ correlates better with $\log T_{\text{eff}}$ (with $r = 0.86$). Surprisingly, they did not study the dependence of the ratio v_∞/v_{esc} with T_{eff} , which is the relation predicted by the radiation-driven wind models.

We found that the ratio v_∞/v_{esc} correlates well with $\log(T_{\text{eff}})$ (Fig. 4), but the correlation is not linear. The ratio is 2.51 ± 0.27 for stars with $\log(T_{\text{eff}}) > 4.40$, 1.28 ± 0.24 at $4.20 < \log(T_{\text{eff}}) < 4.30$, and 0.68 ± 0.14 at $3.90 < \log(T_{\text{eff}}) < 4.0$. There are not enough reliable measurements in the range of $4.00 < \log(T_{\text{eff}}) < 4.20$ to be sure about the behavior of v_∞/v_{esc} in this T_{eff} interval, but our data suggest that there might be a discontinuity near $\log(T_{\text{eff}}) \simeq 4.05$.

We found clear evidence for a discontinuity in v_∞/v_{esc} near $\log(T_{\text{eff}}) \simeq 4.32$ or $T_{\text{eff}} \simeq 21,000$ K at spectral type B1, where the ratio drops by about a factor 2 from 2.5 at the high T_{eff} side to ~ 1.3 at the low T_{eff} side. The values of v_∞ vary from 1200 to 400 km s⁻¹ within one spectral type at B1! We have shown in § 6.1 that this *bistability jump* is related to the degree of ionization in the wind. The stars with high values of v_∞/v_{esc} have a relatively highly ionized wind (strong P Cygni profiles of N v, C iv, and Si iv), whereas the stars with low values of v_∞/v_{esc} have a much lower degree of ionization (strong P Cygni profiles of C ii, Al iii, and Mg ii). This bistability jump is the same as the one found from radiation-driven wind models for the star P Cygni ($T_{\text{eff}} = 19,300$ K) by Pauldrach & Puls (1990). It is due to the rapid change in the degree of ionization in the wind when the wind becomes optically thick in the Lyman continuum (Lamers & Pauldrach 1991).

We suggest that there is probably also a bistability jump in the *mass-loss rates* of supergiants at types B1, in the sense that the stars with lower v_∞ have a higher value of $\dot{M}$ than stars with a higher v_∞ . The correlation is expected to be approximately inversely proportional, with the product $\dot{M}v_\infty$ being constant over the jump, because about the same fraction of the radiative momentum, L/c , is transferred into the wind on both sides of the jump. We have argued in § 6.3 that bistability may be responsible for the formation of disks around B[e]-supergiants.

The terminal velocities were compared with the predictions from the radiation-driven wind theory using the analytical solutions of the line-driven wind equations by Kudritzki et al. (1989; Appendix B) and the force multiplier parameters from Abbott (1982). We found that the values of v_∞ are significantly lower than the predicted ones. This indicates that the values of the force multiplier parameter α were overestimated by Abbott.

We have derived empirical values of α for each star. These are plotted versus $\log(T_{\text{eff}})$ in Figure 7. Their dependence on T_{eff} is described by equations (2) and (3). With these values of α_{emp} and with the equations of Appendix B one can predict the value of v_∞ to an accuracy better than 5% if the stellar parameters are accurately known.

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APPENDIX A

TERMINAL VELOCITIES OF ADDITIONAL STARS

Table 6 gives a list of the terminal velocities measured with an uncertainty larger than 10% and a quality factor of $Q > 4$. Although we did not use these data in this study, they are a useful extension of our data set for other studies of wind velocities.

APPENDIX B

FORMULAE FOR PREDICTING v_∞

The formulae for predicting the terminal velocity of radiation stellar winds are described by Kudritzki et al. (1989). The value of v_∞/v_{esc} depends on the force multiplier parameters α and δ . For $\alpha = \alpha_{\text{emp}}$ (Fig. 7) and $\delta = 0.10$ the predictions agree with our determinations of v_∞ . We list the relevant equations.

$$v_\infty = v_{\text{esc}} \sqrt{\frac{\alpha}{1 - \alpha}} \sqrt{I} \quad (4)$$

TABLE 6
STARS WITH LOWER QUALITY DETERMINATIONS OF v_∞

Atlas number	HD number	Spectral type	T_{eff} (K)	M_{bol}	v_∞ (obs) (km s ⁻¹)	line code (1)	Q (2)
9	46150	O5 V((f))	44300	-9.6	2900 ± 200	2e	7
24	42088	O6.5 V	41200	-8.4	2200 ± 300	2w	8
30	36879	O7 V(n)	40100	-9.3	2000 ± 500	1w,2w	8
31	91824	O7 V((f))	40100	-8.6	2200 ± 300	1d,2e	7
35	152590	O7.5 V	39100	-8.1	2000 ± 300	2w	8
36	152408	O8: Iafpe	34000	-10.3	1200 ± 300	1b,2w	5
40	46966	O8 V	38000	-8.4	2000 ± 200	1d,2e	7
41	48279	O8 V	38000	-8.1	1500	1w,2w	9
43	14633	ON8 V	38000	-8.0	2000 ± 500	1w	8
47	46149	O8.5 V	37000	-8.3	1500	2w	9
51	46202	O9 V	35900	-7.7	1500	2w	9
52	93028	O9 V	35900	-7.9	1500	2w	9
55	201345	ON9 V	35900	-7.7	1300	2w	9
59	209975	O9.5 Ib	30900	-8.8	1800 ± 200	2e,3e	7
60	189957	O9.5 III	32900	-8.7	1500	2w	9
63	149038	O9.7 Iab	30500	-9.7	1800 ± 200	1e,3e	7
68	37128	B0 Iae	26000	-9.4	1800 ± 200	2e,3e	7
70	164402	B0 Ib	26000	-8.8	2000 ± 300	2e,3e	7
72	43818	B0 II	27500	-8.1	1800 ± 300	2e,3e	7
73	48434	B0 III	29000	-7.3	1500 ± 300	1e,2e	7
74	74753	B0 IIIIn	29000	-7.3	1500	2e	9
76	149438	B0 V	30000	-6.5	1000	2w,3w	9
77	38771	B0.5 Ia	23400	-8.8	1400 ± 300	2e,3e	7
78	150898	B0.5 Ia	23400	-9.1	1100 ± 300	1w,2e	7
79	167756	B0.5 Ia	23400	-9.1	1900 ± 300	2e,3e	7
81	91943	B0.5 Ib	23400	-8.3	1500 ± 200	2e,3e	7
96	214080	B1 Ib	20800	-7.7	800	2w,4w	9
97	165793	B1 II	20800	-7.3	1000	2w,3w	9
116	52089	B2 II	18500	-6.4	600	2w,3w	9
131	42087	B2.5 Ibe	17400	-7.6	600 ± 100	2e,3e	7
139	225094	B3 Iae	16300	-8.5	550 ± 200	2w,3e	7
142	53138	B3 Iab	16300	-7.8	550 ± 100	2e,3e	7
148	157038	B4 Ia	14900	-8.1	750 ± 200	3w,5b	5
149	58350	B5 Ia	13600	-7.9	450 ± 100	3e,5e	7
151	36371	B5 Iab	13600	-8.1	400 ± 100	2w,5e	7
152	75129	B5 Ib	13600	-6.3	450 ± 100	3e,5w	7
153	86440	B5 Ib	13600	-6.3	500 ± 100	2e,3e	7
154	164353	B5 Ib	13600	-6.3	500 ± 100	2e,3w	7
155	83183	B5 II	14300	-5.1	500 ± 100	2e,3w	7
156	175156	B5 II	14300	-5.1	450 ± 100	2e,3e	7
161	74371	B6 Iae	12600	-7.8	400 ± 50	2e,3e	7
169	166937	B8 Iape	11200	-8.9	120 ± 30	5b,10d	5
170	199478	B8 Iae	11200	-7.8	300 ± 100	5e,10w	7
173	208501	B8 Ib	11200	-7.0	300 ± 100	3e,10e	7
176	202850	B9 Iab	10300	-6.7	350 ± 50	3w,10w	8
177	212593	B9 Iab	10300	-6.7	350 ± 100	2w,3w	8
196	62623	A2 Iabe	9080	-6.8	150 ± 50	8b,9e	6
198	223385	A3 Iae	8770	-8.7	180 ± 20	9e	7
230	54605	F8 Ia	6100	-8.0	80 ± 20	9b	5

(1) See Table 3.

(2) Quality code as defined in Table 4.

with

$$I = \frac{1-\alpha}{\alpha} \left(\frac{v_c}{v_{\text{esc}}} \right)^2 + Z_0 a_0 \left[u_c - \frac{1}{2} - \frac{\zeta}{3} \left(u_c^3 - \frac{1}{8} \right) \right] - Z_0 a_1 \left[\frac{1}{2} \left(u_c^2 - \frac{1}{4} \right) + \frac{\zeta}{4} \left(u_c^4 - \frac{1}{16} \right) \right] + Z_0 \left[\frac{1}{2} + \frac{1}{8} \frac{\alpha}{1-\alpha} \left(\frac{1}{6} + \frac{\zeta}{80} \right) + \frac{\zeta}{24} \right], \quad (5)$$

with

$$v_c = v_s \sqrt{\frac{\phi}{1-\alpha}}, \quad (6)$$

$$v_s = 9.085 \times 10^3 \left\{ \frac{2 + [1 + I(\text{He})] Y T_{\text{eff}}}{1 + 4Y} \right\}^{1/2}, \quad (7)$$

where $I(\text{He})$ is the ionization fraction of He, $Y = n(\text{He})/n(\text{H})$ and

$$\phi = 3(v_s/v_{\text{esc}})^{0.108 + 0.30 \log(v_s/v_{\text{esc}})}. \quad (8)$$

The functions are

$$\zeta = \frac{q}{1-\alpha}, \quad (9)$$

$$q = 7.5^\delta - 1, \quad (10)$$

$$u_c = 1/x_c, \quad (11)$$

$$x_c = \max\{1.03, x_c(1)\}, \quad (12)$$

$$x_c(1) = \left[\frac{1.46\alpha^{3/5}(\alpha+1)}{\sigma} \right]^{1/3} - \frac{2}{3} \frac{q_c}{\sigma} \quad (13)$$

and

$$\sigma = \left(\frac{v_{\text{esc}}}{v_s} \right)^2 \frac{\alpha^2(1-\alpha)}{\phi^2}. \quad (14)$$

The constant q_c is

$$q_c = \eta^\delta - 1, \quad (15)$$

with $\eta = 22.5$ if $\beta_c = 2$, 7.5 if $\beta_c = 1$, 4.0 if $\beta_c = 0.7$, 2.5 if $\beta_c = 0.5$, and 1.18 if $\beta_c = 0.25$. The values of β_c depend on δ and α . For $\alpha \geq 0.70$, which is of interest for this study, $\beta_c = 2.0$ if $\delta < 0.03$, $\beta_c = 1.0$ if $0.03 \leq \delta < 0.55$, $\beta_c = 0.70$ if $0.055 \leq \delta < 0.085$, $\beta_c = 0.50$ if $0.085 \leq \delta < 0.095$, and $\beta_c = 0.25$ for $\delta \geq 0.095$.

The constants are

$$a_0 = 2 - \left(\frac{1}{1+\alpha} \right)^{1/(1-\alpha)}, \quad (16)$$

$$a_1 = 2a_0 - 2, \quad (17)$$

$$Z_0 = G \left\{ 1 + \left[\frac{2}{\alpha} \left(1 - \frac{1}{G} \right) \right]^{1/2} \right\}, \quad (18)$$

with

$$G = \frac{(qu_c^2 + 1)^{(1/\alpha)-1}}{a_0 - a_1 u_c}. \quad (19)$$

With these equations for $\alpha = \alpha_{\text{emp}}$ and $\delta = 0.10$ the terminal velocity of a radiation-driven stellar wind can be calculated to an accuracy better than 5% if the stellar parameters are known.

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